Tutorial Sheet Simplification

1. Write an expression for Z

A	B	C	**Z**
0	0	0	**0**
0	0	1	**0**
0	1	0	**1**
0	1	1	**0**
1	0	0	**0**
1	0	1	**0**
1	1	0	**1**
1	1	1	**0**

2. Write an expression for Z

A	B	C	**Z**
0	0	0	**1**
0	0	1	**0**
0	1	0	**1**
0	1	1	**0**
1	0	0	**1**
1	0	1	**0**
1	1	0	**1**
1	1	1	**0**

Simplify the expression if possible.

3. Write an expression for Z

A	B	C	D	**Z**
0	0	0	0	**1**
0	0	0	1	**1**
0	0	1	0	**0**
0	0	1	1	**0**
0	1	0	0	**1**
0	1	0	1	**1**
0	1	1	0	**0**
0	1	1	1	**0**
1	0	0	0	**0**
1	0	0	1	**0**
1	0	1	0	**0**
1	0	1	1	**0**
1	1	0	0	**0**
1	1	0	1	**0**
1	1	1	0	**0**
1	1	1	1	**0**

Simplify the expression if possible.

4. Design a circuit which outputs a 1 when input one is opposite to input 3. (In logic value).

5. A majority function of three variables is TRUE when two or more of the variables are TRUE. Create the truth table for the function. Hence, write the equation, simplify if possible and draw the circuit.

6. A security alarm is activated, if movement is detected in a building or if it dark and the personal alarm is pressed. Create the truth table for the function. Hence, write the equation, simplify if possible and draw the circuit.

7. Simplify:

$$Z = AB + B\left(\bar{B}D + A + C(1 + D)\right) + A\bar{A} + A$$

8. Simplify:

Z = (A+B)(A+D)

9. Simplify:

$$Z = (A + B)(A + \bar{B})(A + C)$$